

[Formula]

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$2. \int \frac{dx}{x} = \ln x + c$$

$$3. \int e^x dx = e^x + c$$

$$4. \int e^{mx} dx = \frac{e^{mx}}{m} + c$$

$$5. \int \sin x dx = -\cos x + c$$

$$6. \int \cos x dx = \sin x + c$$

$$7. \int dx = x + c$$

$$8. \int \sec^2 x dx = \tan x + c$$

$$9. \int \sec x \cdot \tan x dx = \sec x + c$$

$$10. \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$11. \int \operatorname{cosec} x \cdot \cot x dx = -\operatorname{cosec} x + c$$

$$12. \int b^x dx = \frac{b^x}{\ln b} + c$$

$$13. \int \tan x dx = \ln|\sec x| + c$$

$$/-\ln|\cos x| + c$$

21.01.2024

MAT120

Integral Calculus and Differential Equation

Marks Dist. ~~5%~~ Attendance = 5%

Assignment = 10% (Best 2)

Quiz = 15% (Best 2)

Mid = 20%

Final = 30%

Lab = 20%

Total = 100%

Class code: nkhybyp

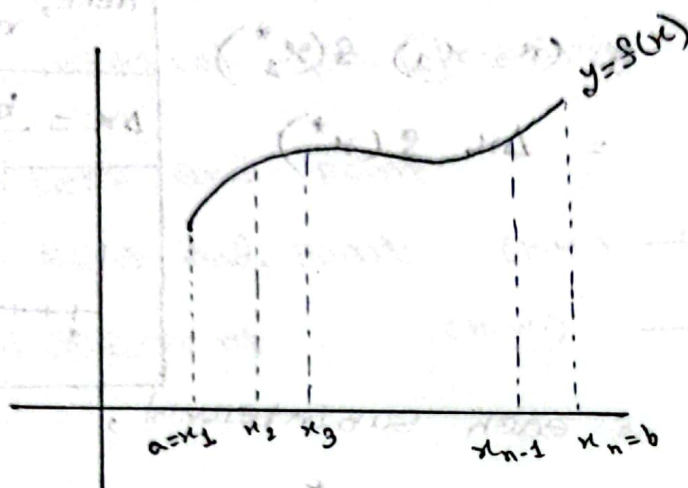
Res:

- ① Calculus with analytic geometry - by
Howard Anton
- ② Calculus, Early Transcendentals (9th ED)
by ~~Dennis G. Zill~~ Stewart, Clegg, Watson
- ③ A First Course in Differential Equations with
modelling Applications (7th ED) - by Dennis G. Zill

Course Outline:

- 2 sets • Integration using Riemann Sums, Indefinite and Definite integrals, Integral with variable limits, Infinite Integration, Integration by substitutions.
- 2 sets • Improper integral, ^{★(V.V.I.)} Integration by parts, Reduction formula, Integration by Trigonometric substitution, Integration of Rational Functions.
- 1 set • Beta gamma function
- 1 set • Length of a curve, surface area of Revolution.

Riemann Sum



Let $y = f(x)$ be a function defined on the closed interval $[a, b]$

(1) Divide the interval $[a, b]$ into n -subintervals

$[x_{k-1}, x_k]$ of width $\Delta x_k = x_k - x_{k-1}$

where, $a = x_1 < x_2 < \dots < x_n = b$

(2) Choose a number x_k^* in each subintervals

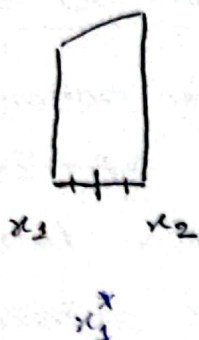
$[x_k - x_{k-1}]$

The n numbers $x_1^*, x_2^*, \dots, x_n^*$ are called sample points.

(3) Calculate the value of $f(x)$ at $x = x_k^*$, then

the sum $A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k \dots \dots (i)$

This is called Riemann Sum



$$A = (x_2 - x_1) f(x_1^*)$$

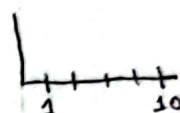
$$= \Delta x f(x^*)$$

4 subinterval $[1, 10]$

Here, $n = 4$ / n

$$\Delta x = \frac{b-a}{n} = \frac{10-1}{4} = \frac{9}{4}$$

$$= \frac{9}{4} / \frac{9}{4}$$



(i) Left end point of each subinterval,

$$x_k^* = a + (k-1)\Delta x$$

(ii) Right end point " " " interval $[a, b]$

$$x_k^* = a + k\Delta x$$

(iii) midpoint

$$x_k^* = a + \left(k - \frac{1}{2}\right) \Delta x$$

$$A \approx \sum_{k=1}^n f(x_k^*) \Delta x$$

Ex - 1

Find the area between the graph of $f(x) = x - 1$ at the interval $[0, 1]$ using

(i) Left end point

(ii) Right end point (H.W.) $\rightarrow A = -\frac{1}{2}$

(iii) Mid point (H.W.) $\rightarrow A = -\frac{1}{2}$

$$\Rightarrow \textcircled{i} \quad A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x \quad \dots \dots \dots (i)$$

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$$

$$\begin{aligned} x_k^* &= a + (k-1) \Delta x \\ &= 0 + (k-1) \frac{1}{n} \\ &= \frac{(k-1)}{n} \end{aligned}$$

$$\begin{aligned} f(x_k^*) &= \frac{k-1}{n} - 1 \\ &= \frac{k-n-1}{n} \end{aligned}$$

$$\therefore A = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k-n-1}{n} \right] \frac{1}{n}$$

$$k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 \quad \left| \begin{array}{l} k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 \\ = \frac{n(n+1)(2n+1)}{6} \end{array} \right| \quad \left| \begin{array}{l} 1+2+3+\dots+n = k \\ = \frac{n(n+1)}{2} \end{array} \right|$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k \cdot n - 1}{n^2} \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k}{n^2} - \frac{1}{n^2} - \frac{1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\sum_{k=1}^n k - \frac{1}{n^2} \sum_{k=1}^n 1 - \frac{1}{n} \sum_{k=1}^n 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \cdot \frac{n(n+1)}{2} - \frac{1}{n^2} \cdot n - \frac{1}{n} \cdot n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{(n+1)}{2n} - \frac{1}{n} - 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n}{2n} + \frac{1}{2n} - \frac{1}{n} - 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} + \frac{1}{2n} - \frac{1}{n} - 1 \right]$$

$$= \frac{1}{2} + 0 - 0 - 1$$

$$= -\frac{1}{2}$$

(Ans)

Ex-2 [when n is given]

Find the Riemann sum of the function $f(x) = 3x+1$ over $[2,6]$ with 4 subintervals.

i) Left end point

ii) Right end point

iii) Midpoint

$$\textcircled{\text{iii}} \Rightarrow \Delta x = \frac{b-a}{n} \quad \left| \quad x_k^* = a + \left(k - \frac{1}{2}\right) \Delta x \right.$$
$$= \frac{6-2}{4}$$
$$= 1$$
$$= 2 + \left(k - \frac{1}{2}\right) \cdot 1$$
$$= 2 + k - \frac{1}{2}$$
$$= \left(k + \frac{3}{2}\right)$$

$$k=1, x_1^* = \frac{5}{2}, f(x_1^*) = \left(3 \times \frac{5}{2} + 1\right) = \left(\frac{15}{2} + 1\right) = \frac{17}{2}$$

$$k=2, x_2^* = \frac{7}{2}, f(x_2^*) = \left(3 \times \frac{7}{2} + 1\right) = \left(\frac{21}{2} + 1\right) = \frac{23}{2}$$

$$k=3, x_3^* = \frac{9}{2}, f(x_3^*) = \left(3 \times \frac{9}{2} + 1\right) = \left(\frac{27}{2} + 1\right) = \frac{29}{2}$$

$$k=4, x_4^* = \frac{11}{2}, f(x_4^*) = \left(3 \times \frac{11}{2} + 1\right) = \left(\frac{33}{2} + 1\right) = \frac{35}{2}$$

$$A = \sum_{k=1}^4 f(x_k^*) \Delta x$$

$$= f(x_1^*) \Delta x + f(x_2^*) \Delta x + f(x_3^*) \Delta x + f(x_4^*) \Delta x$$

$$= 52 \quad (\text{Ans})$$

Ex-1

$$(ii) \quad \begin{array}{l} a=0 \\ b=1 \end{array} \quad \left| \quad \begin{array}{l} \Delta x = \frac{b-a}{n} \\ = \frac{1-0}{n} \\ = \frac{1}{n} \end{array} \right| \quad f(x) = x-1$$

$$\begin{aligned} x_k^* &= a + (k) \Delta x \\ &= 0 + k \times \frac{1}{n} \\ &= \frac{k}{n} \end{aligned} \quad \left| \quad \begin{array}{l} f(x_k^*) = \frac{k}{n} - 1 \\ = \frac{k-n}{n} \end{array} \right.$$

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*)$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k}{n} - 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \frac{k}{n} - \sum_{k=1}^n 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{k=1}^n k - n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \cdot \frac{n(n+1)}{2} - n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n+1}{2} - n \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k-n}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} [k-n]$$

$$\lim_{n \rightarrow \infty} \left[\frac{n^2}{2} + \frac{n}{2} - n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n+1}{2} - n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n}{2} + \frac{1}{2} - n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} - \frac{1}{2} \times n \right]$$

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k-n}{n} \right] \times \left[\frac{1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k-n}{n^2} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sum_{k=1}^n k - \frac{1}{n} \sum_{k=1}^n 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \times \frac{n(n+1)}{2} - \frac{1}{n} \times n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n+1}{2n} - 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} + \frac{1}{2n} - 1 \right]$$

$$= \frac{1}{2} + 0 - 1$$

$$= -\frac{1}{2} \quad (\text{Ans})$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{2} - 1 \right) + 0 = -\frac{1}{2}$$

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(iii)

$$x_k^* = a + \left(k - \frac{1}{2}\right) \Delta x$$

$$= 0 + \left(k - \frac{1}{2}\right) \frac{1}{n} = \frac{2k-1}{2n}$$

$$\Delta x = \frac{b-a}{n} = \frac{1}{n}$$

$$f(x_k^*) = \frac{2k-1}{2n} - 1$$

$$= \frac{2k-1-2n}{2n}$$

$$= \frac{k - \frac{1}{2} - n}{n}$$

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \cdot \Delta x$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{k - \frac{1}{2} - n}{n} \right) \cdot \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k}{n^2} - \frac{1}{2n^2} - \frac{1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sum_{k=1}^n k - \frac{1}{n^2} \sum_{k=1}^n 1 - \frac{1}{n} \sum_{k=1}^n 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \times \frac{n(n+1)}{2} - \frac{1}{n^2} \times \frac{n}{2} - \frac{1}{n} \times n \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n+1}{2n} - \frac{1}{2n} - 1 \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{2} + \frac{1}{8n} - \frac{1}{2n} - 1 \right]$$

$$= \frac{1}{2} + 0 - 0 - 1$$

$$= -\frac{1}{2}$$

Ans

Ex-2

$$i) \Delta x = \frac{b-a}{n}$$

$$= \frac{6-2}{4}$$

$$= 1$$

$$\begin{cases} a = 2 \\ b = 6 \\ n = 4 \end{cases}$$

$$f(x) = 3x + 1$$

$$x_k^* = a + (\Delta x)k \\ = 2 + k$$

$$f(x_k^*) = 3(2+k) + 1 \\ = 6 + 3k + 1 \\ = 3k + 7$$

$$k=1, x_1^* = 2+1 = 3, f(x_1^*) = 3 \times 3 + 1 = 10$$

$$k=2, x_2^* = 2+2 = 4, f(x_2^*) = 3 \times 4 + 1 = 13$$

$$k=3, x_3^* = 2+3 = 5, f(x_3^*) = 3 \times 5 + 1 = 16$$

$$k=4, x_4^* = 2+4 = 6, f(x_4^*) = 3 \times 6 + 1 = 19$$

$$A = \sum_{k=1}^4 f(x_k^*) \Delta x$$

$$= f(x_1^*) \Delta x + f(x_2^*) \Delta x + f(x_3^*) \Delta x + f(x_4^*) \Delta x$$

$$= 10 + 13 + 16 + 19 = 58$$

$$i) \Delta x = \frac{b-a}{n} \\ = \frac{6-2}{4} = 1$$

$$f(x) = 3x + 1$$

$$x_k^* = a + (k-1)\Delta x$$

$$= 2 + (k-1) \times 1$$

$$= 2 + k - 1$$

$$= k + 1$$

$$f(x_k^*) = 3(k+1) + 1$$

$$k=1, x_1^* = 1+1=2, f(x_1^*) = 3 \times 2 + 1 = 7$$

$$k=2, x_2^* = 2+1=3, f(x_2^*) = 3 \times 3 + 1 = 10$$

$$k=3, x_3^* = 3+1=4, f(x_3^*) = 3 \times 4 + 1 = 13$$

$$k=4, x_4^* = 4+1=5, f(x_4^*) = 3 \times 5 + 1 = 16$$

$$A = \sum_{k=1}^4 f(x_k^*) \Delta x$$

$$= f(x_1^*) \Delta x + f(x_2^*) \Delta x + f(x_3^*) \Delta x + f(x_4^*) \Delta x$$

$$= 7 + 10 + 13 + 16$$

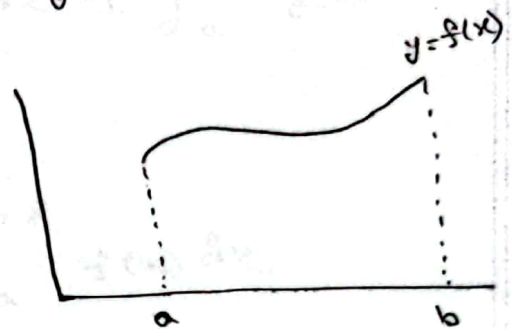
$$= 46$$

(46)

Area under the Curve

Let the area under the curve $y = f(x)$ be

$$A = \int_{x=a}^b f(x) dx$$

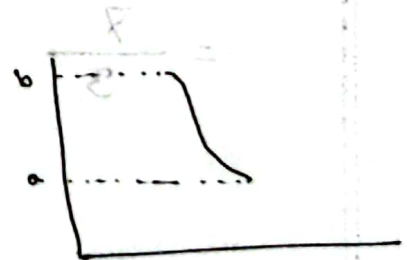


Where, $x = a$ is the lower limit
 $x = b$ is the Upper limit
 $f(x)$ is the integrand

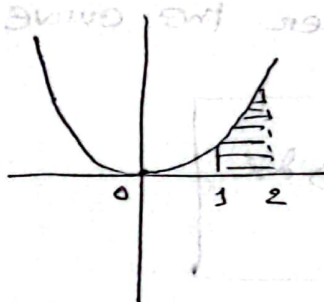
Here,

we integrate the function with respect to x

similarly, $A = \int_{y=a}^b f(y) dy$



Ex-1) Find the area of the curve, $y = x^2$ over the $[1, 2]$



$$A = \int_1^2 x^2 dx$$

$$= \left[\frac{x^3}{3} \right]_1^2$$

$$= \frac{8}{3} - \frac{1}{3}$$

$$= \frac{7}{3}$$

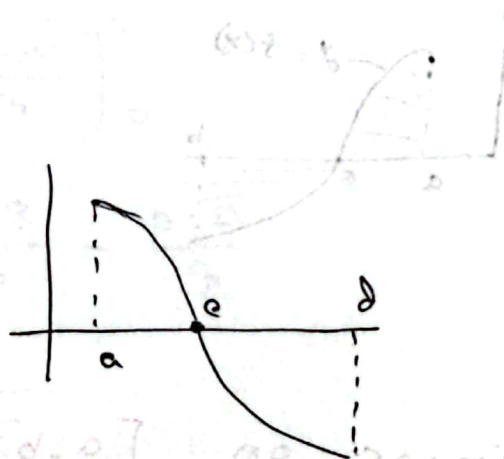
[Some basic properties of integration]

$$1. \int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$2. \int_a^a f(x) dx = 0$$

$$3. \int_a^c f(x) dx + \int_c^d f(x) dx = \int_a^d f(x) dx$$

$$4. \int k dx = k \int dx = kx + C \quad [\because k = \text{constant}]$$



Continuity implies integrability

→ If f is continuous on a closed interval $[a, b]$ then f is said to be integrable on that interval.

like $\int_{-1}^3 \frac{1}{x^2} dx$ is not valid

$$x^2 = 0$$

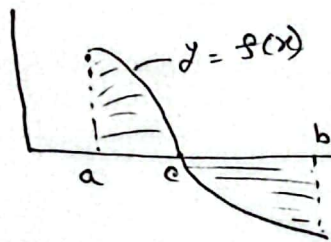
$\therefore x=0 \rightarrow$ in this point there is discontinuity

Net signed Area / Area

$$A = \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

(+ve) (-ve)

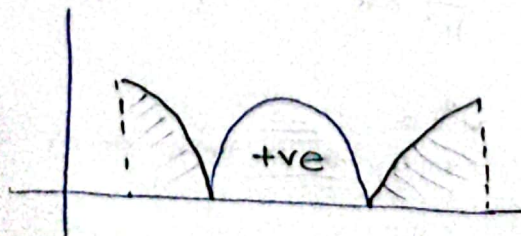
If f is continuous and attain both positive & negative values of f on $[a, b]$



Total Area:

If f is continuous on $[a, b]$ then the total areas,

$$A = \int_a^b |f(x)| dx$$
$$= \int_a^c f(x) dx + \int_c^{c'} f(x) dx + \int_{c'}^b f(x) dx$$



Ex-1:

Find the net signed area and total area of the function $y = 1 - x^2$ over $[0, 2]$

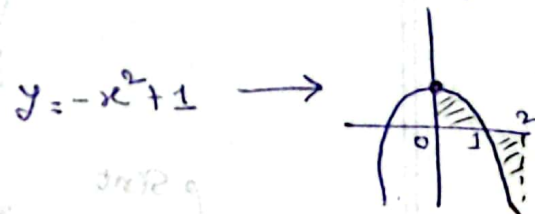
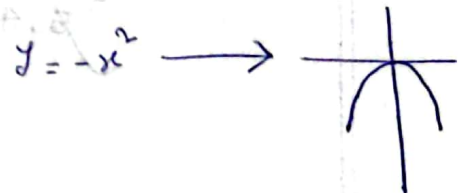
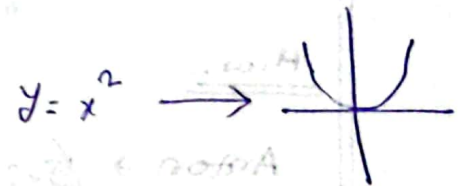
$$A = \int_0^2 (1 - x^2)$$

$$= \left[x - \frac{x^3}{3} \right]_0^2$$

$$= \left[\frac{3x - x^3}{3} \right]_0^2$$

$$= \frac{6 - 8}{3} - \frac{0 - 0}{3}$$

$$= \frac{-2}{3}$$



$$A = \int_0^2 |f(x)| dx$$

$$= \int_0^1 f(x) dx - \int_1^2 f(x) dx$$

$$= \left[\frac{3x - x^3}{3} \right]_0^1 - \left[\frac{3x - x^3}{3} \right]_1^2$$

$$= \left[\frac{3 - 1}{3} \right] - \left[\frac{6 - 8}{3} - \frac{3 - 1}{3} \right]$$

$$= \frac{2}{3} - \left[\frac{-2}{3} + \frac{2}{3} \right]$$

$$= 3 \times \frac{2}{3}$$

$$= 2$$

(Ans)

H.w.

$$\text{Anton} \rightarrow \begin{array}{l} \sqrt{5} \cdot 2 \rightarrow 15 - 34 \\ \sqrt{5} \cdot 4 \rightarrow 27 - 48 \end{array}$$

[Integrals with limit variables]

$$\int_{(t-2)}^{\sin t} (x^2 - 2x + 3) dx$$

$$= \left[\frac{x^3}{3} - \frac{2x^2}{2} + 3x \right]_{(t-2)}^{\sin t}$$

$$= \left[\frac{(\sin t)^3}{3} - \frac{2(\sin t)^2}{2} + 3(\sin t) \right] -$$

$$\left[\frac{(t-2)^3}{3} - \frac{2(t-2)^2}{2} + 3(t-2) \right]$$

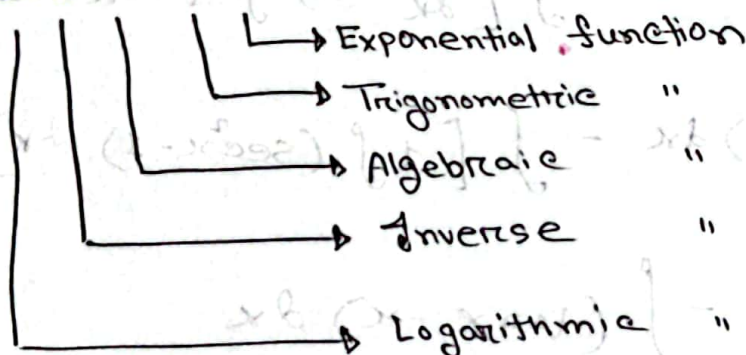
$$= \frac{\sin^3 t}{3} - \sin^2 t + 3 \sin t - \frac{t^3}{3} + 3t^2 - 11t + \frac{38}{3}$$

(Ans)

Integration by parts

$$\int uv dx = u \int v dx - \int \left[\frac{d}{dx}(u) \int v dx \right] dx$$

LIATE



LI
↓ ↓
u v

AE
↓ ↓
u v

Ex-1: $\int x \tan^2 x \, dx$

$u = x$

$v = \tan^2 x$

$$x \int \tan^2 x \, dx - \int \left[\frac{d}{dx}(x) \int \tan^2 x \, dx \right] dx$$

$$= x \int (\sec^2 x - 1) \, dx - \int \left[1 \int (\sec^2 x - 1) \, dx \right] dx$$

$$= x (\tan x - x) - \int (\tan x - x) \, dx$$

$$= x \tan x - x^2 - \ln \sec x + \frac{x^2}{2} + C$$

(ANS)

H.W.

✓ $\int e^x \sin x \, dx$

Anton:

✓ 5.1 → 13-18

✓ 5.2 → 9-33

Integration by Substitutions

Ex-1:

Evaluate, $\int_0^2 x(x^2+1)^3 dx$ — (i)

Let, $x^2+1 = z$

$$\Rightarrow 2x dx = \frac{dz}{dx}$$

$$\Rightarrow 2x dx = dz$$

$$\Rightarrow x dx = \frac{dz}{2}$$

x	0	2
z	1	5

$$(i) \Rightarrow \int_1^5 z^3 \frac{dz}{2}$$

$$= \frac{1}{2} \int_1^5 \frac{z^4}{4}$$

$$= \frac{1}{2} \left[\frac{z^4}{4} \right]_1^5$$

$$= \frac{1}{2} \left[\frac{5^4}{4} - \frac{1}{4} \right]$$

$$= 78$$

1) $\sqrt{f}$

2) $\sqrt[3]{f}$

3) $(f)^2$

4) $(f)^3$

5) $(f)^{1/3}$

Ex-2:

$$\int_0^{\frac{\pi}{4}} \sin^5 2x \cdot \cos 2x \, dx \quad \text{--- (1)}$$

Let,

$$\sin 2x = z$$

$$\Rightarrow 2 \cos 2x = \frac{dz}{dx}$$

$$\Rightarrow dz = 2 \cos 2x \, dx$$

$$\Rightarrow \cos 2x \, dx = \frac{dz}{2}$$

x	0	$\frac{\pi}{4}$
z	0	$\frac{1}{\sqrt{2}}$

$$\textcircled{1} \Rightarrow \int_0^{\frac{1}{\sqrt{2}}} z^5 \cdot \frac{dz}{2}$$

$$= \frac{1}{2} \int_0^{\frac{1}{\sqrt{2}}} z^5 \, dz$$

$$= \frac{1}{2} \left[\frac{z^6}{6} \right]_0^{\frac{1}{\sqrt{2}}}$$

$$= \frac{1}{12} \left[\left(\frac{1}{\sqrt{2}} \right)^6 - 0 \right]$$

$$= \frac{1}{96}$$

Ex-3:

$$\int_2^5 (2x-5)(x-3)^9 dx$$

$$z = x-3$$

$$\Rightarrow dz = dx$$

$$\begin{aligned} 2x-5 &= 2(z+3)-5 \\ &= 2z+6-5 \\ &= 2z+1 \end{aligned}$$

x	2	5
z	-1	2

$$\Rightarrow \int_{-1}^2 z^9 (2z+1) dz$$

$$= \int_{-1}^2 (2z^{10} + z^9) dz$$

$$= \left[\frac{2z^{11}}{11} + \frac{z^{10}}{10} \right]_{-1}^2$$

$$= \frac{52283}{110}$$

(Ans)

H.W. | 1) $\int_0^{\ln 2} e^x (1+e^x)^{1/2} dx$

$$5.9 \rightarrow (5-18)$$

$$7.2 \rightarrow (1-12)$$

7.3 $\rightarrow$ problems related to our lecture

[Reduction Formula] (Important)

Ex: Obtain the reduction formula for

$$\int \sin^n x \, dx$$

$\rightarrow$ H.W.

Hence evaluate $\int \sin^6 x \, dx$

$\tan^n x$
 $\sec^n x$

Solⁿ:

$$\text{Let, } I_n = \int \sin^n x \, dx \quad \text{--- (i)}$$

$$= \int \sin^{n-1} x \cdot \sin x \, dx$$

$$= \sin^{n-1} x \int \sin x \, dx - \int \left[\frac{d}{dx} (\sin^{n-1} x) \int \sin x \, dx \right] dx$$

$$= \sin^{n-1} x (-\cos x) - \int [(n-1) \sin^{n-2} x \cdot \cos x (-\cos x)] dx$$

$$= -\cos x \cdot \sin^{n-1} x + \int [(n-1) \sin^{n-2} x \cdot \cos^2 x] dx$$

$$= -\cos x \cdot \sin^{n-1} x + \int [(n-1) \sin^{n-2} x \cdot (1 - \sin^2 x)] dx$$

$$= -\cos x \cdot \sin^{n-1} x + \int [(n-1) \sin^{n-2} x - (n-1) \sin^n x] dx$$

$$= -\cos x \cdot \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

$$= -\cos x \cdot \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$$

$$\Rightarrow I_n + (n-1) I_n = -\cos x \cdot \sin^{n-1} x + (n-1) I_{n-2}$$

$$\Rightarrow n I_n = -\cos x \cdot \sin^{n-1} x + (n-1) I_{n-2}$$

$$\Rightarrow \boxed{I_n = \frac{-\cos x \cdot \sin^{n-1} x}{n} + \frac{(n-1)}{n} I_{n-2}} \quad (\text{V.V.I.})$$

Therefore,

$$I_6 = -\frac{1}{6} \sin^5 x \cos x + \frac{5}{6} I_4$$

$$= -\frac{1}{6} \sin^5 x \cos x + \frac{5}{6} \left[-\frac{1}{4} \sin^3 x \cos x + \frac{3}{4} I_2 \right]$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{15}{24} I_2$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x$$

$$+ \frac{15}{24} \left[-\frac{1}{2} \sin x \cos x + \frac{1}{2} I_0 \right]$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{5}{8}$$

$$\left[-\frac{1}{2} \sin x \cos x + \frac{1}{2} \int dx \right]$$

$$= -\frac{1}{6} \sin^5 x \cos x - \frac{5}{24} \sin^3 x \cos x + \frac{5}{8} \left[-\frac{1}{2} \sin x \cos x + \frac{1}{2} x \right] + C$$

$\int \sin^n x \, dx$
 $= \int \sin^{n-2} x \cdot \sin x \, dx$

Trigonometric Integrals

Suppose we have to evaluate $\int \sin^m x \cdot \cos^n x \, dx$
where, m, n are integers.

Case-1:

1) when n is odd $\rightarrow$ 1. Split off the term a factor of $\cos x$

2. Apply the identity,
 $\cos^2 x = 1 - \sin^2 x$

3. Make the substitution,
 $z = \sin x$

$$\begin{aligned} & \int \sin^4 x \cos^5 x \, dx \\ \textcircled{1} &= \int \sin^4 x \cdot \cos^4 x \cdot \cos x \, dx \\ &= \int \sin^4 x \cdot (\cos^2 x)^2 \cdot \cos x \, dx \\ \textcircled{2} &= \int \sin^4 x \cdot (1 - \sin^2 x)^2 \cdot \cos x \, dx \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad z &= \sin x \\ dz &= \cos x \, dx \end{aligned}$$

$$\therefore \int \sin^4 x \cdot (1 - \sin^2 x)^2 \cos x \, dx$$

$$= \int z^4 \cdot (1 - z^2)^2 \, dz$$

$$= \int z^4 \cdot (1 - 2z^2 + z^4) \, dz$$

$$= \int (z^4 - 2z^6 + z^8) \, dz$$

$$= \frac{z^5}{5} - \frac{2z^7}{7} + \frac{2z^9}{9} + C$$

$$= \frac{\sin^5 x}{5} - \frac{2\sin^7 x}{7} + \frac{\sin^9 x}{9} + C$$

Case-2:

2) when m is odd $\rightarrow$ 1. Split off the term a factor of $\sin x$.

2. Apply the identity,

$$\sin^2 x = 1 - \cos^2 x$$

3. make the substitution,

$$z = \cos x$$

$$\int \sin^5 x \cos^4 x \, dx$$

$$= \int \sin x \cdot \sin^4 x \cdot \cos^4 x \, dx$$

$$= \int \sin x \cdot (1 - \cos^2 x)^2 \cdot \cos^4 x \, dx.$$

$$z = \cos x$$

$$\Rightarrow dz = -\sin x \, dx$$

$$= - \int (1 - z^2)^2 \cdot z^4 dz$$

$$= - \int (1 - 2z^2 + z^4) \cdot z^4 dz$$

$$= - \int (z^4 - 2z^6 + z^8) dz$$

$$= - \frac{z^5}{5} + \frac{2z^7}{7} + \frac{z^9}{9} + C$$

$$= - \frac{\cos^5 x}{5} + \frac{2\cos^7 x}{7} + \frac{\cos^9 x}{9} + C$$

Case - 3 :

3) when $\left. \begin{array}{l} m \text{ is even} \\ n \text{ is even} \end{array} \right\} \rightarrow$

(1) Use the identity

$$2 \sin^2 x = (1 - \cos 2x)$$

$$\therefore \sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$2 \cos^2 x = (1 + \cos 2x)$$

$$\therefore \cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

(2) Then use the reduction formula

$$\begin{aligned}
 \underline{\text{Ex:}} \quad & \int \sin^4 x \cos^4 x \, dx \\
 &= \int (\sin^2 x)^2 (\cos^2 x)^2 \, dx \\
 &= \int \left[\frac{1}{2} (1 - \cos 2x) \right]^2 \left[\frac{1}{2} (1 + \cos 2x) \right]^2 \, dx \\
 &= \frac{1}{16} \int [(1)^2 - (\cos 2x)^2]^2 \, dx \\
 &= \frac{1}{16} \int [1 - \cos^2 2x]^2 \, dx \\
 &= \frac{1}{16} \int (\sin^2 2x)^2 \, dx \\
 &= \frac{1}{16} \int \sin^4 2x \, dx \quad \text{--- (i)}
 \end{aligned}$$

Let,

$$2x = z$$

$$\Rightarrow 2dx = dz$$

$$\therefore dx = \frac{dz}{2}$$

$$\begin{aligned}
 \text{(i)} \Rightarrow & \frac{1}{16} \int \sin^4 z \cdot \frac{dz}{2} \\
 &= \frac{1}{32} \int \sin^4 z \, dz
 \end{aligned}$$

9

$$I_n = \frac{-\cos x \cdot \sin^{n-1} x}{n} + \frac{(n-1)}{n} I_{n-2}$$

$$\therefore I_4 = \frac{-\cos x \cdot \sin^3 x}{4} + \frac{3}{4} \int \sin^2 x dx$$

$$= -\frac{1}{4} \cos x \cdot \sin^3 x + \frac{3}{4} \left[\frac{-\cos x \sin x}{2} + \frac{1}{2} I_0 \right]$$

$$= -\frac{1}{4} \cos x \cdot \sin^3 x - \frac{3}{8} \cos x \cdot \sin x$$

$$+ \frac{3}{8} x + C$$

(i) $\Rightarrow$

$$\frac{1}{32} \left[-\frac{1}{4} \cos x \cdot \sin^3 x - \frac{3}{8} \cos x \cdot \sin x + \frac{3}{8} x \right] + C$$

Integrating products of Tangents and secants

If m, n are positive integers then the
integral

$\int \tan^m x \cdot \sec^n x \, dx$ can be evaluated

Case-1

If n even $\rightarrow$ (i) Split off a factor of $\sec^2 x$

(ii) Apply $\sec^2 x = 1 + \tan^2 x$

(iii) Make a transformation,
 $z = \tan x$

Case-2

If m odd $\rightarrow$ (i) " " " " ,
 $\sec x \tan x$

(ii) " " $\tan^2 x = \sec^2 x - 1$

(iii) " " " ,

$z = \sec x$

$$\textcircled{1} \Rightarrow \underline{\text{Ex:}} \int \tan^3 x \sec^2 x \, dx \quad \text{--- (i) } \left[\frac{P}{Q} \right] =$$

Let,

$$z = \tan x$$

$$dz = \sec^2 x \, dx$$

$$\textcircled{1} \Rightarrow \int z^3 \cdot dz \quad \text{--- (i) } \left\{ \begin{array}{l} \text{move } z^3 \text{ to } dz \\ \text{then } \int \end{array} \right.$$

$$= \frac{z^4}{4} + C$$

$$= \frac{\tan^4 x}{4} + C$$

(Ans) (ii)

$$\textcircled{2} \Rightarrow \underline{\text{Ex:}} \int \tan^3 x \cdot \sec^2 x \, dx$$

$$= \int \tan^2 x \cdot \sec x \cdot \sec x \cdot \tan x \, dx$$

$$= \int (\sec^2 x - 1) \sec x \cdot \sec x \cdot \tan x \, dx \quad \text{--- (i)}$$

Let,

$$z = \sec x$$

$$dz = \sec x \cdot \tan x \, dx$$

$$\textcircled{1} \Rightarrow \int (z^2 - 1) z \cdot dz$$

$$= \int (z^3 - z) \, dz$$

$$= \left[\frac{x^4}{4} - \frac{x^2}{2} \right]$$

$$= \frac{\sec^4 x}{4} - \frac{\sec^2 x}{2} + C$$

Case-3

If $\left. \begin{array}{l} m \text{ even} \\ n \text{ odd} \end{array} \right\} \rightarrow$ (i) Use $\tan^2 x = \sec^2 x - 1$

to reduce the power of $\sec x$ alone.

(ii) Use the reduction formula for the power of $\sec x$

Ex. $\int \sec^3 x \cdot \tan^2 x \, dx$

$$= \int \sec^3 x \cdot (\sec^2 x - 1) \, dx$$

$$= \int (\sec^5 x - \sec^3 x) \, dx$$

$\Downarrow$
Reduction
Formula

$\Downarrow$
Reduction
Formula

$$\int \sec x \, dx$$

$$= \ln(\sec x + \tan x)$$

$$\int \sec^3 x \, dx$$

$$= \int \sec x \cdot \sec^2 x \, dx$$

$$= \sec x \int \sec^2 x \, dx - \int \left[\frac{d}{dx}(\sec x) \int \sec^2 x \, dx \right] dx$$

$$= \sec x \cdot \tan x \, dx - \int [\sec x \cdot \tan x \cdot \tan x] dx$$

$$= \sec x \cdot \tan x \, dx - \int [\sec x \cdot \tan^2 x] dx$$

$$= \sec x \cdot \tan x \, dx - \int \sec x (\sec^2 x - 1) dx$$

$$= \sec x \cdot \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

$$\Rightarrow \sec x \cdot \tan x$$

$$\Rightarrow 2 \int \sec^3 x \, dx = \sec x \cdot \tan x + \int \sec x \, dx$$

$$\Rightarrow \int \sec^3 x \, dx = \frac{1}{2} [\sec x \cdot \tan x + \ln(\sec x + \tan x)]$$

$$\int \sec^5 x \, dx$$

$$= \int$$

Integral of the form

$$\int \sin mx \cos nx \, dx \quad / \quad \int \sin mx \sin nx \, dx \quad / \quad \int \cos mx \cos nx \, dx$$

then use the identities:

$$\textcircled{1} \sin \alpha \cdot \cos \beta = \frac{1}{2} [\sin(\alpha - \beta) + \sin(\alpha + \beta)]$$

$$\textcircled{2} \sin \alpha \cdot \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\textcircled{3} \cos \alpha \cdot \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

Ex: $\int \sin 7x \cos 4x \, dx$

$$= \int \frac{1}{2} [\sin 3x + \sin 11x] \, dx$$

$$= \frac{1}{2} \left[\frac{-\cos 3x}{3} - \frac{\cos 11x}{11} \right] + C$$

Method of Trigonometric Substitution

Some expression of the form:

$\int \sqrt{a^2 - x^2} \, dx$ $\int \sqrt{a^2 - x^2}$ $= \int \sqrt{a^2 - (a \sin x)^2}$ $= \int \sqrt{a^2 (1 - \sin^2 x)}$ $= \int a \sqrt{\cos^2 x}$ $= \int a \cos x$ $= a \sin x + C$	$\int \sqrt{x^2 + a^2}$ $x = a \tan x$ $\int \sqrt{(a \tan x)^2 + a^2}$ $= \int \sqrt{a^2 (\tan^2 x + 1)}$ $= \int a \sqrt{\sec^2 x}$ $= \int a \sec x$ $= a [\ln(\sec x + \tan x)] + C$	$\int \sqrt{x^2 - a^2}$ $x = a \sec x$ $\int \sqrt{(a \sec x)^2 - a^2}$ $= \int \sqrt{a^2 (\sec^2 x - 1)}$ $= \int \sqrt{a^2 (\tan^2 x)}$ $= \int a \tan x$ $= a \ln \sec x + C$
---	--	--

① $\sin^2 \theta + \cos^2 \theta = 1$
 $\sin \theta = \sqrt{1 - \cos^2 \theta}$
 $\cos \theta = \sqrt{1 - \sin^2 \theta}$

② $\sec^2 \theta - \tan^2 \theta = 1$
 $\sin \theta = \sqrt{1 + \tan^2 \theta}$
 $\tan \theta = \sqrt{\sec^2 \theta - 1}$

"a" is a constant

Ex-

$$\int \frac{dx}{x^2 \sqrt{4 - x^2}}$$

$$= \int \frac{dx}{x^2 \sqrt{2^2 - x^2}}$$

Let, $x = 2 \sin \theta$

$dx = 2 \cos \theta \, d\theta$

$$\int \frac{2 \cos \theta \, d\theta}{4 \sin^2 \theta \sqrt{4 - 4 \sin^2 \theta}}$$

$$= \frac{1}{2} \int \frac{\cos \theta \, d\theta}{\sin^2 \theta \sqrt{4(1 - \sin^2 \theta)}}$$

$$= \frac{1}{4} \int \frac{\cos \theta \, d\theta}{\sin^2 \theta \cdot \cos \theta}$$

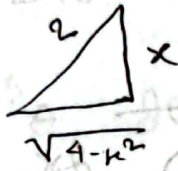
$$= \frac{1}{4} \int \frac{1}{\sin^2 \theta} \, d\theta$$

$$= -\frac{1}{4} \cot \theta \quad \text{--- (ii)}$$

$$x = 2 \sin \theta$$

$$\Rightarrow \sin \theta = \frac{x}{2}$$

$$\therefore \cot \theta = \frac{\sqrt{4-x^2}}{x}$$



$$(ii) \Rightarrow -\frac{1}{4} \times \frac{\sqrt{4-x^2}}{x} + C$$

Ans

H.W.

$$\int \frac{\sqrt{x^2-25}}{x} dx$$

$$x = 5 \sec \theta$$

Square root of Quadratics

Ex: $\int \sqrt{4x^2 - 16x + 52} \, dx$

$$= \int \sqrt{4(x^2 - 4x + 13)} \, dx$$

$$= 2 \int \sqrt{x^2 - 4x + 13} \, dx$$

$$= 2 \int \sqrt{(x-2)^2 + 9} \, dx$$

$$= 2 \int \sqrt{(x-2)^2 + (3)^2} \, dx \quad \text{--- (i)}$$

let,

$$x-2 = z$$

$$\Rightarrow dx = dz$$

$$(i) \Rightarrow 2 \int \sqrt{z^2 + 3^2} \, dz$$

$$= 2 \int \sqrt{(3 \tan \theta)^2 + 3^2} \, dz$$

$$\dots 3 \sec^2 \theta \cdot d\theta$$

$$= 2 \int \sqrt{9 \tan^2 \theta + 9} \cdot 3 \sec^2 \theta \, d\theta$$

(ii)
let,

$$z = 3 \tan \theta$$

$$\Rightarrow 3 \sec^2 \theta \cdot d\theta = dz$$

$$= 2 \int 3 \sqrt{\tan^2 \theta + 1} \cdot 3 \sec^2 \theta \, d\theta$$

$$= 18 \int \sec \theta \cdot \sec^2 \theta \, d\theta$$

$$= 18 \int \sec^3 \theta \, d\theta$$

Reduction
Formula

$$= 18 \times \frac{1}{2} [\sec \theta \cdot \tan \theta + \ln(\sec \theta + \tan \theta)]$$

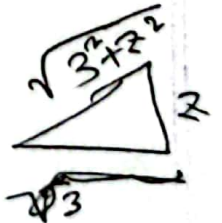
$$= 9 [\sec \theta \cdot \tan \theta + \ln(\sec \theta + \tan \theta)]$$

$$= 9 \left[\frac{\sqrt{9+2^2}}{3} \cdot \frac{2}{3} \right]$$

$$+ \ln \left(\frac{\sqrt{9+2^2}}{3} + \frac{2}{3} \right) + C$$

$$\Rightarrow z = 3 \tan \theta$$

$$\Rightarrow \tan \theta = \frac{2}{3}$$



$$= 18 \left[\frac{1}{18} (x-2) \sqrt{(x-2)^2 + 9} \right]$$

$$+ \frac{1}{2} \ln \left(\frac{\sqrt{(x-2)^2 + 9}}{3} + \frac{(x-2)}{3} \right) + C$$

(Ans)

Partial Fraction

$$\textcircled{1} \frac{1}{(x-a)(x-b)} = \frac{A}{(x-a)} + \frac{B}{(x-b)}$$

$$\textcircled{2} \frac{1}{(x-a)^2 + (x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$$

$$\textcircled{3} \frac{1}{(x-a)(ax^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+c}{ax^2+bx+c}$$

$$\textcircled{4} \frac{1}{(x-a)(ax^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+c}{ax^2+bx+c}$$

$$2 + \left[\frac{B}{x} + \frac{C}{x^2} + \frac{Dx+E}{(ax^2+bx+c)^2} \right]$$

9th / 10th

Higher mathematics

Ch - 2 $\Rightarrow$ Partial Fraction part

[Practice]

Integration by substitutions

$$1) \int_0^{\ln 2} e^x (1+e^x)^{\frac{1}{2}} dx$$

$$\text{let, } 1+e^x = z$$

$$\Rightarrow e^x dx = dz$$

x	0	ln 2
z	2	3

$$\therefore \int_2^3 z^{\frac{1}{2}} dz$$

$$= \int_2^3 z^{\frac{1}{2}} dz$$

$$= \left[\frac{z^{\frac{3}{2}}}{\frac{3}{2}} \right]_2^3$$

$$= \frac{2}{3} \cdot 1.58$$

Ans

$$\int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$\int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx$$

$$\int \sec^n x \, dx = \frac{\sec^{n-2} x (\tan x)}{n-1} + \frac{(n-2) \int \sec^{n-2} x \, dx}{n-1}$$

$$\frac{d}{dx} \left[\sin^{-1} \left(\frac{x}{a} \right) \right] = \frac{1}{\sqrt{a^2 - x^2}}$$

$$\frac{d}{dx} \left[\frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) \right] = \frac{1}{a^2 + x^2}$$

$$\frac{d}{dx} \left[\cos^{-1} \left(\frac{x}{a} \right) \right] = \frac{-1}{\sqrt{a^2 - x^2}}$$

$$\frac{d}{dx} \left(\frac{1}{a} \cot^{-1} \left(\frac{x}{a} \right) \right) = \frac{-1}{a^2 + x^2}$$

$$\frac{d}{dx} \left[\frac{1}{a} \sec^{-1} \left| \frac{x}{a} \right| \right] = \frac{1}{x \sqrt{x^2 - a^2}}$$

$$\frac{d}{dx} \left[\frac{1}{a} \operatorname{cosec}^{-1} \left| \frac{x}{a} \right| \right] = \frac{-1}{x \sqrt{x^2 - a^2}}$$

Integrating Rational Fractions by Partial Fraction:

$$\frac{Q(x)}{P(x)} = \frac{A}{P-x} + \frac{B}{(P-x)(L(x))} + \dots$$

Linear factor Rule:

$$\frac{1}{(ax+b)^m} = \frac{A_1}{(ax+b)} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_m}{(ax+b)^m}$$

$$\frac{1}{(x-1)^3} = \frac{A_1}{(x-1)} + \frac{B_1}{(x-1)^2} + \frac{C_1}{(x-1)^3}$$

Ex-

$$\int \frac{5x-10}{x^2-3x-4} dx \quad \text{(i)}$$

$$\frac{5x-10}{x^2-3x-4} = \frac{5x-10}{x^2-4x+x-4} = \frac{5x-10}{(x+1)(x-4)} = \frac{A}{x-4} + \frac{B}{x+1} \quad \text{(ii)}$$

$$\Rightarrow (5x-10) = (x+1)A + B(x-4) \quad \text{(ii)}$$

$$x = -1, \text{ (ii)} \Rightarrow A = -15$$

$$B = 3$$

$$x = 4, \text{ (i)} \Rightarrow A = 2$$

(iii) $\Rightarrow$

$$\frac{5x-10}{(x+1)(x-4)} = \frac{2}{x-4} + \frac{3}{x+1}$$

(i) $\Rightarrow$

$$\int \left[\frac{2}{x-4} + \frac{3}{x+1} \right] dx$$

$$= 2 \ln(x-4) + 3 \ln(x+1) + C$$

(Ans)

Quadratic Factor

$$\int \frac{x^2+x-2}{3x^3-x^2+3x-1} dx$$

$$= \int \frac{x^2+x-2}{x^2(3x-1) + 1(3x-1)} dx$$

$$= \int \frac{x^2+x-2}{(3x-1)(x^2+1)} dx$$

$$\frac{x^2+x-2}{(3x-1)(x^2+1)} = \frac{A}{(3x-1)} + \frac{Bx+C}{(x^2+1)}$$

$$\Rightarrow x^2+x-2 = A(x^2+1) + (Bx+C)(3x-1)$$

$$\Rightarrow x^2+x-2 = Ax^2+A+3Bx^2-Bx+3Cx-C$$

$$\Rightarrow x^2+x-2 = x^2(A+3B) + x(3C-B) + A-C$$

Equating the coefficients of x^2 , x and constant term,

$$A+3B=1$$

$$3C-B=1$$

$$A-C=-2$$

$$\left\{ \begin{array}{l} A = -\frac{7}{5} \\ B = \frac{4}{5} \\ C = \frac{3}{5} \end{array} \right.$$

~~$$x^2+x-2 = x^2 + x - 2$$~~

$$\int \left[\frac{-\frac{7}{5}}{(3x-1)} + \frac{\frac{4}{5}x + \frac{3}{5}}{(x^2+1)} \right] dx$$

$$= -\frac{7}{5} \int \frac{dx}{3x-1} + \frac{4}{5} \int \frac{x dx}{x^2+1} + \frac{3}{5} \int \frac{1}{1+x^2} dx$$

$$= -\frac{7}{5} \ln(3x-1) \times \frac{1}{3} + \frac{2}{5} \int \frac{2x}{x^2+1} dx + \frac{3}{5} \int \frac{1}{1+x^2} dx$$

$$\int \frac{f(x)}{f(x)} dx = \ln|x| + C$$

$$= \frac{-7}{15} \ln(3x-1) + \frac{2}{5} \ln(x^2+1) + \frac{3}{5} \tan^{-1} x + C$$



$$\frac{(x-1)}{(x-1)^2} \leftarrow \text{নিম্নের অঙ্কের degree হকি তাই}$$

PROPER FRACTION

$$\frac{(x-1)^2}{(x-1)} \leftarrow \text{Improper fraction}$$

Ex - 1

1) $\int \frac{2x}{x^2 + x - 2} dx$

2) $\int \frac{2x+1}{x^3 - 2x^2} dx$

3) $\int \frac{3x^4 + 4x^3 + 16x^2 - 20x + 9}{(x+2)(x^2+3)^2} dx$

Rational

Improper fraction

$$5) \frac{8}{3} (1$$

$$\frac{8}{5} = 1 + \frac{3}{5}$$

$$= \frac{5+3}{5}$$

$$= \frac{8}{5}$$

Ex- $\int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{(x^2 + x - 2)} dx$ (i)

$$\begin{aligned} & 3x^4 + 3x^3 - 5x^2 + x - 1 \\ &= 3x^4 + 3x^3 - 6x^2 + x^2 + x - 2 + 1 \\ &= 3x^2(x^2 + x - 2) + 1(x^2 + x - 2) + 1 \end{aligned}$$

$$\frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} = \frac{3x^2(x^2 + x - 2)}{x^2 + x - 2} + \frac{x^2 + x - 2}{x^2 + x - 2} + \frac{1}{x^2 + x - 2}$$

$$= 3x^2 + 1 + \frac{1}{x^2 + x - 2}$$

$$= (3x^2 + 1) + \frac{1}{(x+2)(x-1)}$$

$$\begin{aligned} & x^2 + 2x - 1 \\ &= (x+2)(x-1) \end{aligned}$$

$$= 3x^2 + 1 + \left(\frac{A}{x+2} + \frac{B}{x-1} \right)$$

~~$$3x^2 + 3x - 5x^2 + x - 1 = A(x^2 + x - 2) + B$$~~

$$= 3x^2 + 1 + \frac{-\frac{1}{3}}{x+2} + \frac{\frac{1}{3}}{x-1}$$

$$\int (3x^2 + 1 - \frac{1}{3} \times \frac{1}{x+2} + \frac{1}{3} \times \frac{1}{x-1}) dx$$

$$= 3 \frac{x^3}{3} + x - \frac{1}{3} \ln(x+2) + \frac{1}{3} \ln(x-1) + C$$

n.w

7.5 $\Rightarrow$ Ex-1-34

Infinite Integral

If $f(x)$ is a function which is continuous for $x > a$ then we define the infinite integral by $[a, \infty)$

$$\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

If the limit exists and is finite then we can see that the infinite integral ~~converges~~ converges, otherwise diverges.

Similarly, if $f(x)$ is continuous for $x \leq b$

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

Converge $\longrightarrow$ Finite value

Diverge $\longrightarrow$ No fixed value

$$\frac{x^{-1}}{-1} = \frac{x^{-2+1}}{-1+1} = x^{-1}$$

Ex: Test the convergence of the function

$$\int_1^{\infty} \frac{1}{x^2} dx$$

$$\Rightarrow \int_1^{\infty} \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx$$

$$= \lim_{t \rightarrow \infty} \left[\frac{-1}{x} \right]_1^t$$

$$= \lim_{t \rightarrow \infty} \left[-\frac{1}{t} + \frac{1}{1} \right]$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{1}{t} + 1 \right)$$

$$= 0 + 1$$

$$= 1$$

$\therefore$ The function $f(x)$ is convergent at $(1, \infty)$

(An)

$$\int u v dx = u \int v dx - \int \left[\frac{d}{dx}(u) \int v dx \right] dx$$

[Reduction Formula]

$$1) \int \sin^n x dx$$

$$2) \int \tan^n x dx$$

$$3) \int \sec^n x dx$$

①

$$I_n = \int \sin^n x dx$$

$$= \int \sin^{n-1} x \cdot \sin x dx$$

$$= \sin^{n-1} x \int \sin x dx - \int \left[\frac{d}{dx} (\sin^{n-1} x) \int \sin x dx \right] dx$$

$$= \sin^{n-1} x (-\cos x) - \int [(n-1) \sin^{n-2} x (\cos x) \cancel{dx} \cdot (-\cos x)] dx$$

$$= \sin^{n-1} x (-\cos x) + \int [(n-1) \sin^{n-2} x \cdot \cos^2 x] dx$$

$$= \sin^{n-1} x (-\cos x) + \int [(n-1) \sin^{n-2} x \cdot (1 - \sin^2 x)] dx$$

$$= \sin^{n-1} x \cdot (-\cos x) + \int [(n-1) \sin^{n-2} x - (n-1) \sin^n x] dx$$

$$= -\cos x \cdot \sin^{n-1} x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx$$

$$= -\cos x \cdot \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$$

$$\Rightarrow I_n + (n-1) I_n = -\cos x \cdot \sin^{n-1} x + (n-1) I_{n-2}$$

$$\Rightarrow I_n = \frac{-\cos x \cdot \sin^{n-1} x + (n-1) I_{n-2}}{n}$$

(Ans)

2)

$$I_n = \int \tan^n x \, dx$$

$$= \int \tan^{n-2} x \cdot \tan^2 x \, dx$$

~~$$= \tan^{n-2} \int (\tan^2 x) dx = \int \left[\frac{d}{dx} (\tan^{n-2} x) \int \tan^2 x \, dx \right] dx$$~~

$$= \int \tan^{n-2} x (\sec^2 x - 1) \, dx$$

$$= \int (\sec^2 x \cdot \tan^{n-2} x - \tan^{n-2} x) \, dx$$

$$= \int \sec^2 x \cdot \tan^{n-2} x \, dx - \int \tan^{n-2} x \, dx$$

$$= \int \sec^2 x \cdot \tan^{n-2} x \, dx - I_{n-2}$$

Let, $t = \tan x$

$$\Rightarrow \frac{dt}{dx} = \sec^2 x \Rightarrow \sec^2 x \, dx = dt$$

$$\therefore \int t^{n-2} \, dt$$

$$= \frac{t^{n-1}}{n-1} = \frac{\tan^{n-1} x}{n-1}$$

$$\boxed{\therefore I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}}$$

$$I_n = \int \sec^n x \, dx$$

$$= \int \sec^{n-2} x \cdot \sec^2 x \, dx$$

$$= \sec^{n-2} x \int \sec^2 x \, dx - \int \left[\frac{d}{dx} (\sec^{n-2} x) \int \sec^2 x \, dx \right] dx$$

$$= \sec^{n-2} x \cdot \tan x - \int (n-2) \sec^{n-3} x \cdot (\sec x \cdot \tan x) \cdot \tan x \, dx$$

$$= \sec^{n-2} x \cdot \tan x - (n-2) \int \sec^{n-2} x \cdot \tan^2 x \, dx$$

$$= \sec^{n-2} x \cdot \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) \, dx$$

$$= \sec^{n-2} x \cdot \tan x - (n-2) \int \sec^n x \, dx + (n-2) \int \sec^{n-2} x \, dx$$

$$= \sec^{n-2} x \cdot \tan x - (n-2) I_n + (n-2) I_{n-2}$$

$$\Rightarrow I_n + (n-2) I_n = \sec^{n-2} x (\tan x) + (n-2) I_{n-2}$$

$$\Rightarrow I_n (n-1) = \sec^{n-2} x (\tan x) + (n-2) I_{n-2}$$

$$\therefore I_n = \frac{\sec^{n-2} x (\tan x)}{n-1} + \frac{(n-2) I_{n-2}}{n-1}$$

Improper Integrals

Consider the definite integral ~~or any or both~~
 $\int_a^b f(x) dx$ if the integrand f has point of
discontinuity on $[a, b]$ or any or both
 a & b are infinite, then the integral is called
improper integral.

Type-1: $\int_1^{\infty} \frac{dx}{x^2}$, $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$, $\int_{-\infty}^1 e^x dx$ $\rightarrow$ These are
improper
integrals
with infinite
intervals.

Type-2: $\int_{-1}^1 \frac{dx}{x^2}$; $\int_1^2 \frac{dx}{x-1}$ $\rightarrow$ These are improper
integral with
infinite discontinuity

Type-3: $\int_0^{\infty} \frac{dx}{\sqrt{x}}$, $\rightarrow$ This is the improper
integral with infinite
discontinuity
& infinite intervals.

Ex(1)

$$\int_1^{\infty} \frac{dx}{x}$$

$$= \lim_{t \rightarrow \infty} \int_1^t \frac{dx}{x}$$

$$= \lim_{t \rightarrow \infty} [\ln x]_1^t$$

$$= \lim_{t \rightarrow \infty} [\ln t - \ln 1]$$

$$= \infty$$

$\therefore$ is diverge

(Ans)

Ex(3)

$$\int_{-\infty}^1 e^x dx$$

$$= \int_{-\infty}^1 e^x dx$$

$$= \lim_{t \rightarrow -\infty} [e^x]_t^1$$

$$= \lim_{t \rightarrow -\infty} [e^1 - e^t]$$

$$= e^1 - e^{-\infty}$$

$$= e$$

$\therefore$ is diverge

Ex(2)

$$\int_2^{\infty} \frac{dx}{(x-1)^2}$$

$$= \int_2^{\infty} (x-1)^{-2} dx$$

$$= \lim_{t \rightarrow \infty} \left[\frac{(x-1)^{-1}}{-1} \right]_2^t$$

$$= \lim_{t \rightarrow \infty} \left[\frac{1}{-(t-1)} \right]$$

$$+ \frac{1}{1(2-1)}$$

$$= \lim_{t \rightarrow \infty} \left[\frac{1}{1} - \frac{1}{1(t-1)} \right]$$

$$= \frac{1}{1} - 0$$

$$= \frac{1}{1} = 1$$

(Ans)

$\therefore$ is converge

Ex-4

$$\int_{-\infty}^{\infty} \frac{e^{-t}}{1+e^{-2t}} dt \longrightarrow$$

$$\int_{-\infty}^{\infty} \frac{e^{-t}}{1+e^{-2t}} dt \quad \text{--- (i)}$$

$$e^{-t} = z$$

$$\Rightarrow -e^{-t} dt = dz$$

$$\Rightarrow dt = \frac{dz}{-e^{-t}}$$

$$\Rightarrow e^{-t} dt = -dz$$

$$\textcircled{1} \Rightarrow \int \frac{-dz}{1+z^2} \\ = -\tan^{-1} z$$

$$\therefore \int_{-\infty}^{\infty} \frac{e^{-t}}{1+e^{-2t}} dt$$

$$= \int_{-\infty}^0 \frac{e^{-t}}{1+e^{-2t}} dt +$$

$$+ \int_0^{\infty} \frac{e^{-t}}{1+e^{-2t}} dt$$

$$= \int_{-\infty}^0 \frac{e^{-t}}{1+e^{-2t}} dt +$$

$$\int_0^{\infty} \frac{e^{-t}}{1+e^{-2t}} dt$$

$$= \lim_{t \rightarrow -\infty} \left[-\tan^{-1} z \right]_0^0$$

$$+ \lim_{t \rightarrow \infty} \left[-\tan^{-1} z \right]_0^t$$

$$e^{\infty} =$$

$$\frac{1}{e^{\infty}} = e^{-\infty}$$

$$= \lim_{t \rightarrow -\infty} \left[-\tan^{-1} e^{-t} \right]_t^0 + \lim_{t \rightarrow \infty} \left[-\tan^{-1} e^{-t} \right]_0^t$$

$$= -\cancel{\tan^{-1}} e^{-0} + \tan^{-1} e^{\infty} - \tan^{-1} e^{-\infty} + \cancel{\tan^{-1}} e^{-0}$$

$$= -\frac{\pi}{4} + \cancel{\frac{\pi}{4}} + \frac{\pi}{2} - 0 + \frac{\pi}{4}$$

$$= 0$$

$\therefore f$ is converge

Ex

$$\int_1^2 \frac{dx}{1-x}$$

$$= \lim_{t \rightarrow 1} \left[-\ln(1-x) \right]_1^2$$

$$= -\ln(-1) + \ln|1-t|$$

~~$$= -\ln(1) + \ln|1-t|$$~~

~~$$= 0 + \ln|1-t|$$~~

$\therefore$ is diverge

$$\underline{Ex} : \int_1^4 \frac{dx}{(x-2)^{2/3}} = \int_1^2 \frac{dx}{(x-2)^{2/3}} + \int_2^4 \frac{dx}{(x-2)^{2/3}}$$

$$\lim_{t \rightarrow 2} \left[(x-2)^{-2/3} dx \right]_1^2 + \lim_{t \rightarrow 2} \left[\right]$$

$$= \int_1^4 \frac{(x-2)^{1/3}}{\frac{1}{3}} dx$$

$$= \int_1^4 3(x-2)^{1/3} dx$$

$$= \lim_{t \rightarrow 2} \left[3(x-2)^{4/3} \right]_1^t + \lim_{t \rightarrow 2} \left[3(x-2)^{4/3} \right]_t^4$$

$$= 3(t-2)^{4/3} - 3(1-2)^{4/3} + 3(4-2)^{4/3} - 3(t-2)^{4/3}$$

$$= 0 + 3 + 3 \cdot 2^{4/3} - 0$$

$$= 6.75$$

∴ is converge.

$$-\frac{1}{x} \quad \frac{x^{-2+1}}{-2+1} \quad \frac{x^{-1}}{-1}$$

Ex:

$$\int_{-2}^2 \frac{dx}{x^2}$$

$$= \int_{-2}^0 \frac{dx}{x^2} + \int_0^2 \frac{dx}{x^2}$$

$$= \lim_{t \rightarrow 0} \left[-\frac{1}{x} \right]_{-2}^{-t} + \lim_{t \rightarrow 0} \left[-\frac{1}{x} \right]_t^2$$

$$= -\frac{1}{-2} - \frac{1}{-t} \quad \infty \quad -\frac{1}{t} - \frac{1}{2}$$

$$= -\infty + \infty$$

... is ~~converge~~ Diverge

H.W

1) $\int_0^1 \frac{dx}{\sqrt{1-x}}$

2) $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

3) $\int_0^{\infty} (1-x)e^{-x} dx$

[Gamma Beta Function]

The gamma function Γ is defined by the infinite interval.

$$\Gamma_n = \int_0^{\infty} e^{-x} \cdot x^{n-1} dx$$

Some properties:

$$\Gamma_{n+1} = n!$$

$$\Gamma_{n+1} = n \Gamma_n$$

$$\Gamma_1 = 1$$

$$\frac{\Gamma_4}{\Gamma_{3+1}} = 3!$$

(I.) $\Gamma_{\frac{1}{2}} = \sqrt{\pi}$

The beta function $\beta(m, n)$ is defined by the definite integral,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, \quad m, n > 0$$

Some properties:

$$1. \beta(m, n) = \frac{\Gamma_m \cdot \Gamma_n}{\Gamma_{m+n}}$$

(N.U.I.) $2. \int_0^{\pi/2} \sin^p \theta \cdot \cos^q \theta d\theta = \frac{\Gamma_{\frac{p+1}{2}} \Gamma_{\frac{q+1}{2}}}{2 \Gamma_{\frac{p+q+2}{2}}}$

$$3. \Gamma_p \Gamma_{1-p} = \frac{\pi}{\sin p \pi}$$

Ex

$$\int_0^{\pi/2} \sin^3 \theta \cos^4 \theta \, d\theta$$

$$= \frac{\sqrt{\frac{3+1}{2}} \sqrt{\frac{4+1}{2}}}{2 \sqrt{\frac{3+4+2}{2}}} = \frac{\sqrt{2} \sqrt{\frac{5}{2}}}{2 \sqrt{\frac{9}{2}}}$$

$$= \frac{\frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\frac{1}{2}}}{\cancel{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \cancel{\frac{3}{2}} \cdot \frac{1}{2} \cdot \sqrt{\frac{2}{2}}}$$

$$= \frac{2}{35}$$

(Ans)

$$\sqrt{9/2} = \sqrt{\frac{7}{2} + 1}$$

$$= \frac{7}{2} \sqrt{\frac{7}{2}}$$

$$= \frac{7}{2} \sqrt{\frac{5}{2} + 1}$$

$$= \frac{7}{2} \cdot \frac{5}{2} \sqrt{\frac{5}{2}}$$

$$= \frac{7}{2} \cdot \frac{5}{2} \cdot \sqrt{\frac{3}{2} + 1}$$

$$= \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \sqrt{\frac{3}{2}}$$

$$= \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \sqrt{\frac{1}{2} + 1}$$

$$= \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}}$$

Ex -

$$\int_0^{\pi/6} \sin^2 6\theta \cdot \cos^4 3\theta \, d\theta \quad \text{--- (1)}$$

$$z = 3\theta$$

$$\Rightarrow dz = 3 d\theta$$

0	0	$\pi/6$
z	0	$\pi/2$

$$= \frac{1}{3} \int_0^{\pi/2} \sin^2 z \cdot \cos^4 z \, dz$$

$$= \frac{1}{3} \int_0^{\pi/2} (2 \sin z \cdot \cos z)^2 \cdot \cos^4 z \, dz$$

$$= \frac{1}{3} \int_0^{\pi/2} 4 \sin^2 z \cdot \cos^2 z \cdot \cos^4 z \, dz$$

$$= \frac{1}{3} \int_0^{\pi/2} 4 \sin^2 z \cdot \cos^6 z \, dz$$

$$= \frac{1}{3} \left[\frac{4 \sqrt{\frac{2+1}{2}} \sqrt{\frac{6+1}{2}}}{2 \sqrt{\frac{2+6+2}{2}}} \right]$$

$$= \frac{1}{3} \left[\frac{4 \sqrt{\frac{3}{2}} \cdot \sqrt{\frac{7}{2}}}{2 \sqrt{5}} \right]$$

$$= \frac{1}{3} \left[\frac{4 \cdot \cancel{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}} \cdot \cancel{\frac{5}{2}} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}}}{2 \times 4!} \right]$$

$$= \frac{1}{3} \left[\frac{4 \cdot \cancel{\frac{1}{2}} \cdot \frac{1}{2} \cdot \sqrt{\pi} \cdot \cancel{\frac{5}{2}} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{2 \times 4!} \right]$$

$$= \frac{5\pi}{192}$$

Ans

Ex-

$$\int_0^1 \frac{dx}{(1-x^3)^{2/3}} \quad \text{--- (i)}$$

$$z = x^3$$

$$\Rightarrow 3x^2 dx = dz$$

x	0	1
z	0	1

$$\textcircled{1} \Rightarrow \int_0^1 \frac{dz}{3x^2} \cdot \frac{1}{(1-z)^{2/3}}$$

$$= \int_0^1 \frac{dz}{3 \cdot (z^{1/3})^2} \cdot \frac{1}{(1-z)^{2/3}}$$

$$= \frac{1}{3} \cdot \int_0^1 z^{-2/3} (1-z)^{-2/3} dz$$

$$= \frac{1}{3} \int_0^1 z^{\frac{1}{3}-1} (1-z)^{\frac{2}{3}-1} dz$$

$$= \frac{1}{3} B\left(\frac{1}{3}, \frac{2}{3}\right)$$

(Ans)

$$= \frac{1}{3} \cdot \frac{\sqrt{\frac{1}{3}} \sqrt{\frac{2}{3}}}{\sqrt{\frac{1}{3} + \frac{2}{3}}}$$

$$= \frac{1}{3} \cdot \frac{\sqrt{\frac{1}{3}} \cdot \sqrt{1 - \frac{1}{3}}}{\sqrt{1}}$$

$$= \frac{1}{3} \cdot \frac{\sin \frac{1}{3} \pi}{1}$$

$$= \frac{1}{3} \cdot \frac{\pi}{\sin \frac{\pi}{3}}$$

Ans

Ex: $\int_0^{\infty} x^{\frac{1}{4}} \cdot e^{-\sqrt{x}} dx$ — (1)

$$z = \sqrt{x}$$

$$\Rightarrow dz = \frac{1}{2\sqrt{x}} dx$$

x	0	∞
z	0	∞

$$\therefore \int_0^{\infty} e^{-z} \cdot z^{\frac{1}{2}} \cdot 2z dz$$

$$= \frac{3}{2} \sqrt{\pi}$$

$$\frac{\Gamma(\frac{3}{2})}{\Gamma(\frac{3}{2})} = \frac{\Gamma(\frac{1}{2})}{\Gamma(\frac{1}{2})} = 1$$

$$\frac{\Gamma(\frac{3}{2})}{\Gamma(\frac{3}{2})} = \frac{\Gamma(\frac{1}{2})}{\Gamma(\frac{1}{2})} = 1$$

$$z = \sqrt{x}$$

$$dz = \frac{1}{2\sqrt{x}} dx$$

$$\int_0^{\infty} e^{-z} \cdot z^{\frac{1}{2}} \cdot 2z dz$$

Prove that,



Ex:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \int_0^1 x^{\frac{1}{2}-1} (1-x)^{\frac{1}{2}-1} dx$$

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \int_0^1 \frac{1}{\sqrt{x}} \cdot \frac{1}{\sqrt{(1-x)}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{x-x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{-(x^2-x)}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{-(x^2-2x \cdot \frac{1}{2} + (\frac{1}{2})^2 - \frac{1}{4})}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{-(x-\frac{1}{2})^2 - \frac{1}{4}}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{\frac{1}{4} - (x-\frac{1}{2})^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2}} dx$$

$$= \left[\sin^{-1} \frac{x - \frac{1}{2}}{\frac{1}{2}} \right]_0^1$$

$$= \frac{\pi}{2} + \frac{\pi}{2}$$

$$= \pi$$

$$\therefore \beta\left(\frac{1}{2}, \frac{1}{2}\right) = \pi$$

$$\Rightarrow \frac{\sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}}}{\sqrt{\frac{1}{2} + \frac{1}{2}}} = \pi$$

$$\Rightarrow \frac{\left(\sqrt{\frac{1}{2}}\right)^2}{\sqrt{1}} = \pi$$

$$\therefore \sqrt{\frac{1}{2}} = \sqrt{\pi}$$

[Proved]

$$\# \int_0^a y^3 \sqrt{a^4 - y^4} dy \quad \text{--- (1)}$$

$$= \int_0^a \left(1 - \frac{y^4}{a^4}\right)^{1/2}$$

let, $\frac{y^4}{a^4} = z$ $\rightarrow y = \left(a^4 z\right)^{1/4}$

$\therefore a^4 z = y^4$
 $\Rightarrow 4y^3 dz = a^4 dz$
 ~~$\therefore a^4 (1 - z)$~~

y	0	a
z	0	1

$$\therefore \int_0^1 \left(a^4 z\right)^{3/4} \sqrt{a^4 (1-z)} \cdot \frac{a^4 \cdot dz}{4y^3}$$

$$= \int_0^1 a^7 \cdot z^{3/4} \cdot a^2 \sqrt{(1-z)} \cdot \frac{a^4 \cdot dz}{4 \left\{ (a^4 z)^{3/4} \right\}}$$

$$= \frac{a^7 \cdot a^2 \cdot a^4}{4 \cdot a^3} \int_0^1 z^{3/4} \cdot (1-z)^{1/2} dz$$

$$= \frac{a^{10}}{4} \int_0^1 z^{3/4} (1-z)^{1/2} dz$$

$$\int \frac{dx}{\sqrt{(5x)^2 - (2)^2}}$$

$$5x = 2 \sec \theta$$

$$\Rightarrow x = \frac{2 \sec \theta}{5}$$

$$\Rightarrow dx = \frac{2 \sec \theta \cdot \tan \theta \cdot d\theta}{5}$$

$$\therefore \frac{1}{5} \int \frac{2 \sec \theta \cdot \tan \theta \cdot d\theta}{\sqrt{(2 \sec \theta)^2 - 2^2}}$$

$$= \frac{1}{5} \int \frac{2 \sec \theta \cdot \tan \theta \cdot d\theta}{\sqrt{4(\sec^2 \theta - 1)}}$$

$$= \frac{1}{5} \int \frac{2 \sec \theta \cdot \tan \theta \cdot d\theta}{2 \tan \theta}$$

$$= \frac{1}{5} \int \sec \theta \, d\theta$$

$$= \frac{1}{5} \ln | \tan \theta + \sec \theta |$$

$$= \frac{1}{5} \ln \left| \tan \left(\sec^{-1} \frac{5x}{2} \right) + \sec \left(\sec^{-1} \frac{5x}{2} \right) \right|$$

$$= \frac{1}{5} \ln$$

$$\frac{\sin}{\cos} \times \sec^{-1}$$

$$\frac{\sin}{\frac{1}{\sec}} \times \sec^2$$

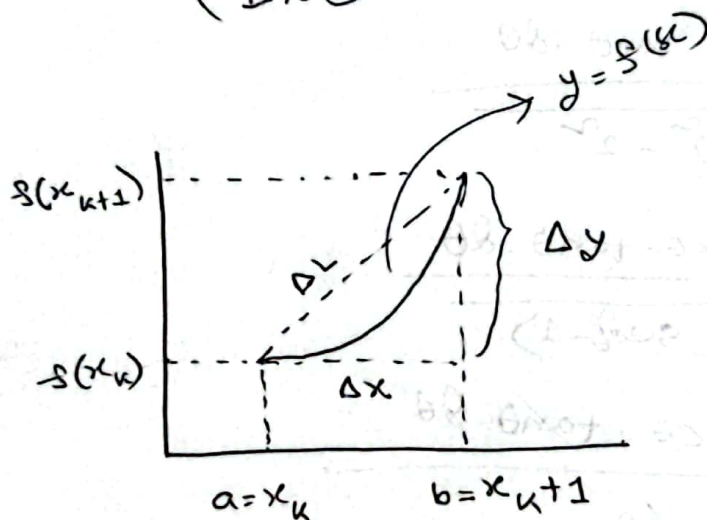
$$\sin \cdot \sec \cdot \sec^2$$

Arc length of a curve

$$(\Delta L)^2 = (\Delta x)^2 + (\Delta y)^2$$

$$\Rightarrow (\Delta L)^2 = (\Delta x)^2 + \left[1 + \left(\frac{\Delta y}{\Delta x}\right)^2\right]$$

$$\Rightarrow \left(\frac{\Delta L}{\Delta x}\right)^2 = 1 + \left(\frac{\Delta y}{\Delta x}\right)^2$$



$$\lim_{\Delta x \rightarrow 0} \frac{\Delta L}{\Delta x} = \lim_{\Delta x \rightarrow 0} \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}$$

$$\Rightarrow dL = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\Rightarrow \int dL = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\therefore L = \int_a^b \sqrt{1 + \{s'(x)\}^2} dx$$

Thus, if $y = s(x)$ is a smooth curve on the interval $[a, b]$, then the arc length L of $s(x)$ over $[a, b]$ is defined by

$$L = \int_{x=a}^b \sqrt{1 + \{s'(x)\}^2} \cdot dx$$

Similarly,

$$L = \int_{y=a}^b \sqrt{1 + \{s(y)\}^2} \cdot dy$$

Ex. 1

Find the arc length of the curve $x = \frac{1}{3}(y^2+2)^{3/2}$
from $y=0$ to $y=1$

$$\begin{aligned} \left(\frac{dx}{dy} \right)^2 &= \left[\frac{d}{dy} \left(\frac{1}{3} (y^2+2)^{3/2} \right) \right]^2 \\ &= \left[\frac{1}{3} \left[\frac{d}{dy} (y^2+2)^{3/2} \right] \right]^2 \\ &= \left[\frac{1}{3} \times \frac{3}{2} \times (y^2+2)^{1/2} \times 2y \right]^2 \\ &= (y (y^2+2)^{1/2})^2 = y^2 \cdot (y^2+2) \end{aligned}$$

$$\int_0^1 \sqrt{1 + y^2 + 2y^2} dy$$

$$= \int_0^1 \sqrt{1 + y^2 + 2y^2} dy$$

$$= \int_0^1 \sqrt{(y^2+1)^2} dy$$

$$= \int_0^1 (y^2+1) dy$$

$$= \left[\frac{y^3}{3} + y \right]_0^1$$

$$= \frac{1}{3} + 1$$

$$= \frac{4}{3}$$

Ans

Ex-2

$$f(x) = \sqrt{x} \quad , \text{ over } [1, 4]$$

$$\frac{d}{dx} (\sqrt{x})$$

$$= \frac{1}{2\sqrt{x}}$$

$$\left(\frac{d}{dx} \right)^2 = \frac{1}{4x}$$

$$\therefore \int_1^4 \sqrt{1 + \frac{1}{4x}} \, dx$$

$$= \int_1^4 \sqrt{1 + \frac{1}{4x}} \, dx$$

$$= \int_1^4 \left(1 + \frac{1}{4x} \right)^{1/2} dx$$

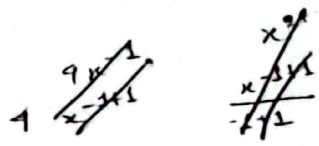
$$\frac{1}{6} + 2 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{6} + \frac{1}{2\sqrt{x}}$$

$$\frac{1x+1}{1x}$$

$$\left(\frac{1}{2} \right)^2 + 2 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}}$$

$$\frac{1}{4} + 2 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}}$$

$$\frac{1}{6} + 2$$



$$= \left[\frac{\left(1 + \frac{1}{4x}\right)^{3/2}}{\frac{3}{2} \left(0 - \frac{1}{4x^2}\right)} \right]_1^4$$

$$= \left[\frac{\left(1 + \frac{1}{4x}\right)^{3/2}}{\frac{3}{2} \left(-\frac{1}{4x^2}\right)} \right]_1^4$$

=

Ex-13:

$$24yx = y^4 + 48, \quad y = 2, 4$$

$$\Rightarrow x = \frac{y^4 + 48}{24y}$$

$$\Rightarrow x = \frac{1}{24} y^3 + \frac{2}{y}$$

$$\therefore \frac{dx}{dy} = \frac{3}{24} y^2 - \frac{2}{y^2}$$

$$\therefore \left(\frac{dx}{dy}\right)^2 = \left(\frac{3}{24} y^2 - \frac{2}{y^2}\right)^2$$

-63.86
-92.46
2.16

25.1
-25.2

$$\therefore \int_2^4 \sqrt{1 + \left(\frac{1}{8} y^2 - \frac{2}{y^2} \right)^2} dy$$

$$= \int_2^4 \sqrt{1 + \frac{y^4}{64} - 2 \cdot \frac{1}{8} y^2 \cdot \left(-\frac{2}{y^2} \right) + \frac{4}{y^4}} dy$$

$$= \int_2^4 \sqrt{1 + \frac{y^4}{64} - \frac{1}{2} + \frac{4}{y^4}} dy$$

$$= \int_2^4 \sqrt{\frac{y^4}{64} + \frac{1}{2} + \frac{4}{y^4}} dy$$

$$= \int_2^4 \sqrt{\left(\frac{y^2}{8} \right)^2 + 2 \cdot \frac{y^2}{8} \cdot \frac{2}{y^2} + \left(\frac{2}{y^2} \right)^2} dy$$

$$= \int_2^4 \sqrt{\left(\frac{y^2}{8} + \frac{2}{y^2} \right)^2} dy$$

$$= \int_2^4 \left(\frac{y^2}{8} + \frac{2}{y^2} \right) dy$$

$$= \left[\frac{y^3}{24} - \frac{2}{y} \right]_2^4 = \frac{13}{6} - \left(-\frac{2}{3} \right) = \frac{17}{6}$$

$$\frac{1}{2} + \frac{2}{4} = \frac{5}{4}$$

Ans

Ex-4 : $y = x^{\frac{2}{3}}$, $x = 1, 8$

$$\frac{dy}{dx} = \frac{2}{3} x^{\frac{2}{3}-1}$$

$$= \frac{2}{3} x^{-\frac{1}{3}}$$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{2}{3} x^{-\frac{1}{3}}\right)^2$$

$$= \frac{4}{9} x^{-\frac{2}{3}}$$

$$\therefore \int_1^8 \sqrt{1 + \frac{4}{9} x^{-\frac{2}{3}}} dx$$

$$= \int_1^8 \sqrt{\cancel{1+2+1} + \frac{4}{9x^{2/3}}} dx$$

$$= \int_1^8 \sqrt{\frac{9x^{2/3} + 4}{9x^{2/3}}} dx$$

$$= \int_1^8 \frac{\sqrt{9x^{2/3} + 4}}{3x^{1/3}} dx$$

7.63

Let,

$$9x^{2/3} + 4 = 2$$

$$\Rightarrow \frac{2}{3} \times 9 \times x^{-1/3} dx = dz$$

$$\Rightarrow 6x^{-1/3} dx = dz$$

$$\Rightarrow \frac{dx}{6x^{1/3}} = \frac{dz}{6}$$

$$\therefore \int_{13}^{40} \frac{\sqrt{z}}{18} dz$$

$$= \frac{1}{18} \int_{13}^{40} \sqrt{z} dz$$

$$= \frac{1}{18} \times \frac{2}{3} \left[z^{3/2} \right]_{13}^{40}$$

$$= 7.63$$

(Ans)

Now.

$$1) y = \frac{x^6 + 8}{16x^2}, \quad x = 2, 3$$

$$2) x = \frac{y^4}{8} + \frac{1}{4y^2}, \quad y = 1, 4$$

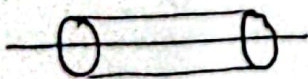
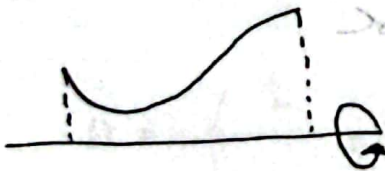
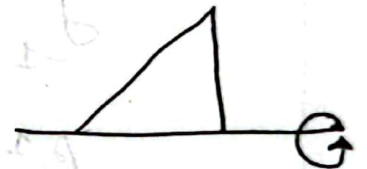
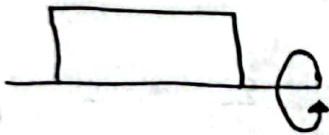
$$3) y = x^{3/2}, \quad y = 1, 2\sqrt{2}$$

$$4) y = x^{3/2}, \quad x = 1, 2$$

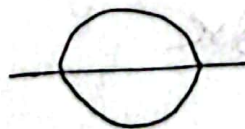
x	1	8
z	13	40

Solids of Revolution

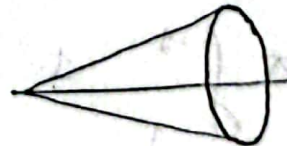
A solid of revolution is a solid that is generated by revolving a plane region about a line that lies in the same plane. This line is called the axis of revolution.



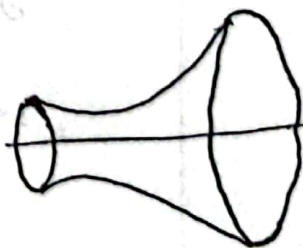
Cylinder



sphere



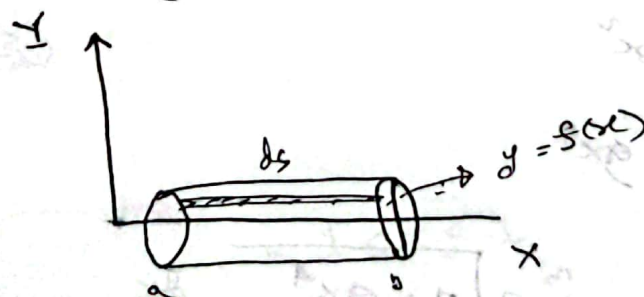
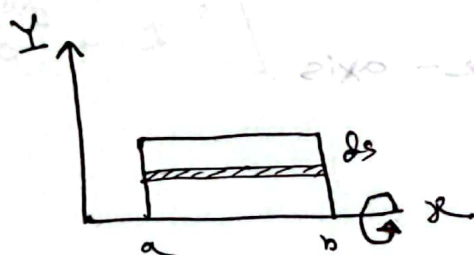
cone



Surface Area:

A surface of revolution is a surface that is generated by revolving a plane curve about an axis.

When the curve is revolved about x-axis



Then, Surface Area:

$$dA = 2\pi r ds$$

$$= 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\int dA = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\therefore A = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$

about
y-axis

$$A = \int_a^b 2\pi s(y) \sqrt{1 + \{s'(y)\}^2} dy$$

Ex-1

Find the area of the surface that is generated by revolving the part of the curve, $y = x^3$ between $x=0$ and $x=1$ about the x -axis.

$$y = x^3$$

$$\Rightarrow \frac{dy}{dx} = 3x^2$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = 9x^4$$

$$\int_0^1 2\pi x^3 \sqrt{1 + 9x^4} dx$$

$$\int_0^1 2\pi x^3 \sqrt{(3x^2)^2 + 2 \cdot 3x^2 \cdot \frac{1}{3x^2} + 1} dx$$

let,

$$1 + 9x^4 = z$$

$$\Rightarrow 36x^3 \cdot dx = dz$$

x	0	1
z	1	10

$$= \frac{2\pi}{36} \int_1^{10} \sqrt{z} \, dz$$

$$= \frac{2\pi}{18} \left[\frac{1}{2\sqrt{z}} \right]_1^{10}$$

$$= \frac{2\pi}{3 \times 18} \left[z^{3/2} \right]_1^{10}$$

$$= \frac{\pi}{27} \left[10^{3/2} - 1^{3/2} \right]$$

$$= 3.563$$

Ans

Ex-2

$$f(x) = \sqrt{4-x^2}, \quad -1 \leq x \leq 1$$

$$\Rightarrow y = \sqrt{4-x^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{4-x^2}} \cdot (-2x)$$

$$= \frac{-x}{\sqrt{4-x^2}}$$

$$= \frac{-x}{\sqrt{4-x^2}}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{x^2}{4-x^2}$$

$$A = \int_{-1}^1 2\pi \sqrt{4-x^2} \cdot \sqrt{1 + \frac{x^2}{4-x^2}} dx$$

$$= \int_{-1}^1 2\pi \sqrt{4-x^2} \cdot \frac{\sqrt{4}}{\sqrt{4-x^2}} dx$$

$$= \int_{-1}^1 2\pi \cdot 2 dx$$

$$= 4\pi \int_{-1}^1 1 dx$$

$$= 4\pi \left[x \right]_{-1}^1$$

$$= 4\pi [1+1]$$

$$= 8\pi$$

$$= 8\pi$$

Ans

H.W.

1) $g(x) = 7x$, $0 \leq x \leq 1$

2) $g(x) = \sqrt{x}$, $1 \leq x \leq 4$

3) $x = 9y + 1$, $0 \leq y \leq 2$

4) $x = \sqrt{9 - y^2}$, $-2 \leq y \leq 2$

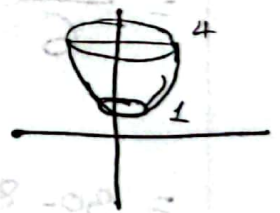
5) $y = x^2$, $1 \leq x \leq 2$, about y axis ($1 \leq y \leq 4$)

6) $f(x) = \frac{1}{2}x^{1/2} - \frac{1}{2}x^{3/2}$,
 $1 \leq x \leq 3$

$1 + (f'(x))^2 = \left(\frac{1}{4}x^{-1/2} + \frac{3}{4}x^{1/2} \right)^2$

$x = \sqrt{y}$

$$\int_1^4 2\pi \sqrt{y} \sqrt{1 + (g'(y))^2} dy$$



$$= \int_1^4 2\pi \sqrt{y} \sqrt{1 + \left(\frac{1}{2\sqrt{y}}\right)^2} dy$$

$$= \int_1^4 2\pi \sqrt{y} \sqrt{1 + \frac{1}{4y}} dy$$

$$= \int_1^4 2\pi \sqrt{y} \sqrt{\frac{4y+1}{4y}} dy$$

$$= \int_1^4 2\pi \sqrt{y} \frac{\sqrt{4y+1}}{2\sqrt{y}} dy$$

$$= \int_1^4 \pi \sqrt{4y+1} \, dy$$

$$= \pi \int_1^4 \sqrt{4y+1} \, dy$$

$$= \pi \times \frac{2}{3} \left[4(4y+1)^{3/2} \right]_1^4$$

$$= \frac{2\pi}{3} \left[4(4 \times 4 + 1)^{3/2} - (4 + 1)^{3/2} \right]$$

$$= \frac{\pi}{6} (17^{3/2} - 5^{3/2})$$

$$= 30.846$$

Ans

$$1) x = \frac{y^4}{4} + \frac{1}{8y^3}, \quad 1 \leq y \leq 2$$

$$2) y = 4 + 3x^2, \quad 1 \leq x \leq 2, \quad \text{about } y \text{ axis}$$

$$3) y = \frac{1}{4} \sqrt{6x+2}, \quad -\frac{1}{\sqrt{2}} \leq y \leq \frac{\sqrt{5}}{2}, \quad x \text{ axis}$$

$$4) y = e^{2y}, \quad -1 \leq y \leq 0, \quad y\text{-axis}$$